

Tanuj Kumar

[Email](#)

[Website](#) | [Google Scholar](#)

BITS Pilani Integrated Dual Degree, 2015 - 2020

B.E. (Hons.) **Electrical and Electronics** & M.Sc. (Hons.) **Chemistry** | CGPA: 7.96/10 (3.57/4.0)

MAJOR PROJECTS

Indian Institute of Science

Sep 2020 - Ongoing | Bengaluru, Karnataka, India | [Prof. Sushobhan Avasthi](#)

Fabricating **quantum dot IR photodetectors** with **cleanroom** processing, **atomic layer deposition**, RF **sputtering** and **thermal evaporation**.

Nano Lab at Tufts University

Aug 2019 - Aug 2020 | Boston, MA, USA | [Prof. Sameer Sonkusale](#)

Thesis on fabrication of flexible **Thread-Based Transistors** (TBT) with ionogels. Developed a **high-throughput** cleanroom-free **fabrication method** to **increase** TBT production by **100 times**.

CentraleSupélec

May 2019 - Jul 2019 | Paris, France | Prof. Caroline Lelandais-Perrault

Designed an asynchronous **Neuromorphic ADC** for use in **ECG monitoring**; Developed a new mathematical model for its figure of merit. **Discovered trend** of temporally decreasing maximum error.

On-Board Computer of a satellite

January 2016 - December 2018 | [Team Anant, BITS Pilani](#)

Designed and implemented a **Linux** based On-Board Computer system, its **drivers**, and **sensor interfaces** for a **nanosatellite**. Built the team's [website](#).

PUBLICATIONS AND TALKS

T. Kumar, R. E. Owyung, S. Sonkusale, "High Throughput Cleanroom-Free Fabrication of Thread Based Transistors", **Materials Research Society**, Nov 2020 (Accepted)

S.M. Raje, A. Goel, S. Sharma, K. Aggarwal, D. Mantri, **T. Kumar**, "Development of On Board Computer for a Nanosatellite", **International Astronautical Congress**, Sep 2017 ([PDF](#))

T. Kumar, R.E. Owyung, S Sonkusale, "Rapid Cleanroom-Free Fabrication of Thread Based Transistors Using Three-Dimensional Stencil-Based Patterning", Submitted

"Building a Satellite Isn't Rocket Science", **TEDx BITS Pilani**, India, Jan 2019 ([Link](#))

"Role of Open Source in the Development of a Student-Built Nanosat & its Team Members", Open Source Cubesat Workshop - **European Space Agency**, Madrid, Sep 2018 ([Link](#))

TEACHING EXPERIENCE

Teaching Assistant - Microelectronic Circuits (EEE F244)

January 2019 - May 2019 | [Prof. Anu Gupta](#)

Taught microelectronic circuit theory and SPICE & Cadence simulation techniques.

Teaching Assistant - Computer Programming (CS F111)

August 2018 - December 2018 | [Prof. Sundaresan Raman](#)

Taught Linux commands and C programming to freshman year students.

COLLEGE PROJECTS

Differential Power Attack Immune Circuits Against Hardware Hacking (Proposal submitted)

August 2018 - May 2019 | [Prof. Anu Gupta](#)

Started the Differential Power Attack project with **Prof. Anu Gupta** at BITS Pilani. **Designed** a dynamically scaled, differential pull-down **circuit** with **~0.03%** current variation. **Authored** our group's **project proposal** to the **Department of Science & Technology (DST)**, Government of India.

Electronic properties of materials in radiation environments

January 2019 - May 2019 | [Dr. Mrinmoyee Basu](#)

Simulated the effects of radiation typically found in the **Large Hadron Collider** on **SiC** and **GaN** MOSFETs in **Sentaurus** TCAD.

Indira Gandhi Center for Atomic Research

May 2017 - Jul 2017 | Kalpakkam, Tamil Nadu, India

Developed a DMA & Interrupts based **nuclear reactor health monitoring** system.

Indian Space Research Organization

May 2018 - Jul 2018 | Bengaluru, Karnataka, India

Designed a real-time **target prediction system** and its **mathematical model** using an **Extended Kalman Filter**. Developed a software camera model as the **testing framework**.

TECHNICAL PROFICIENCY

Lab Skills

Clean room fabrication, deposition, fume hood work, laser cutting, microscopy, Raman, AFM, SEM, Semiconductor analysis

CAD and IDE

MATLAB, Cadence tools, LTSpice, Vivado, LabVIEW, EagleCAD, AutoCAD, Solidworks

Programming

Verilog, C, C++, Assembly, Java, Python, Git, Linux